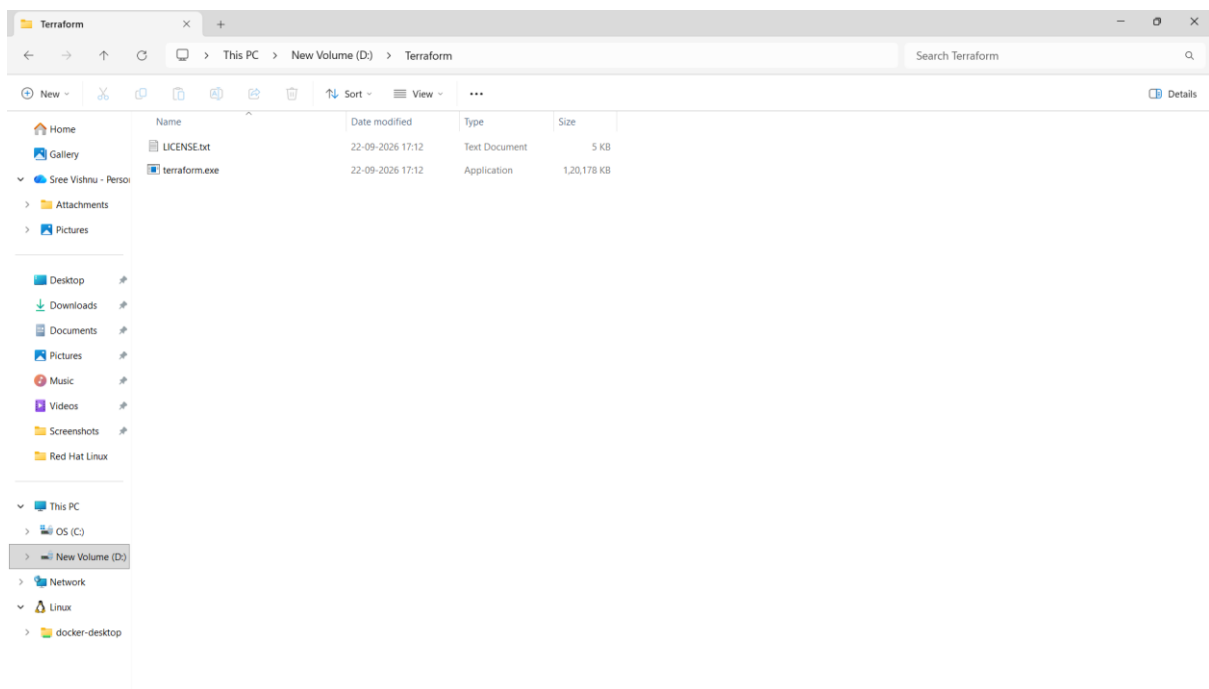
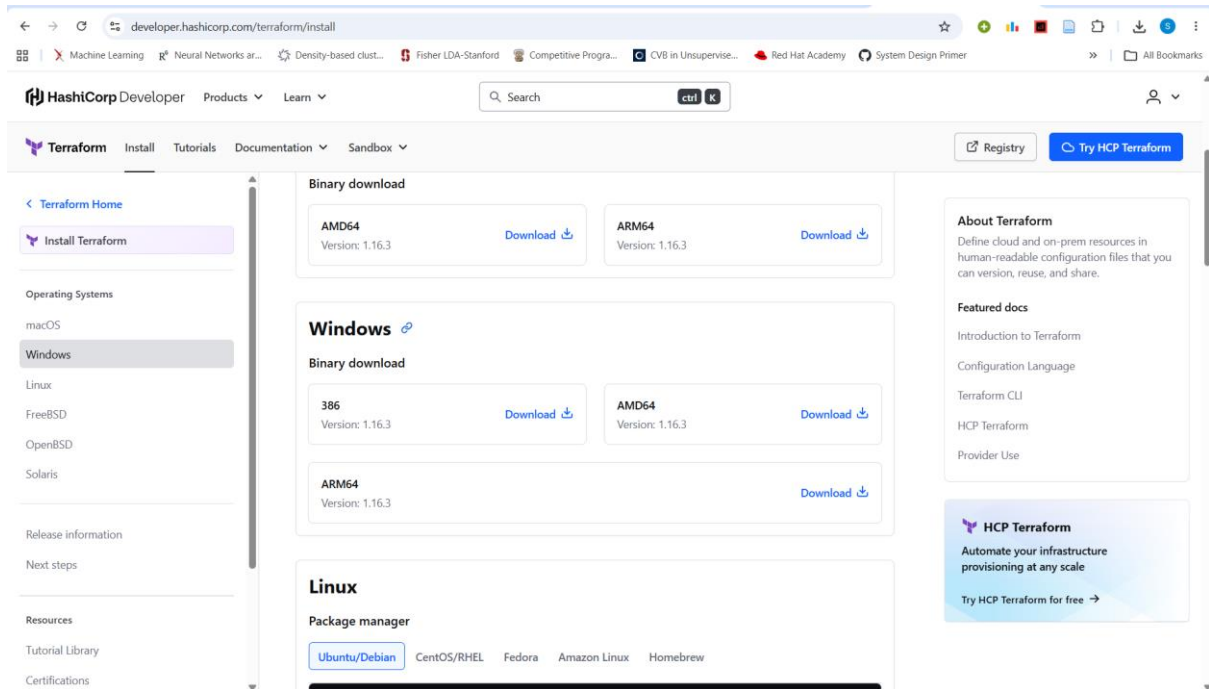
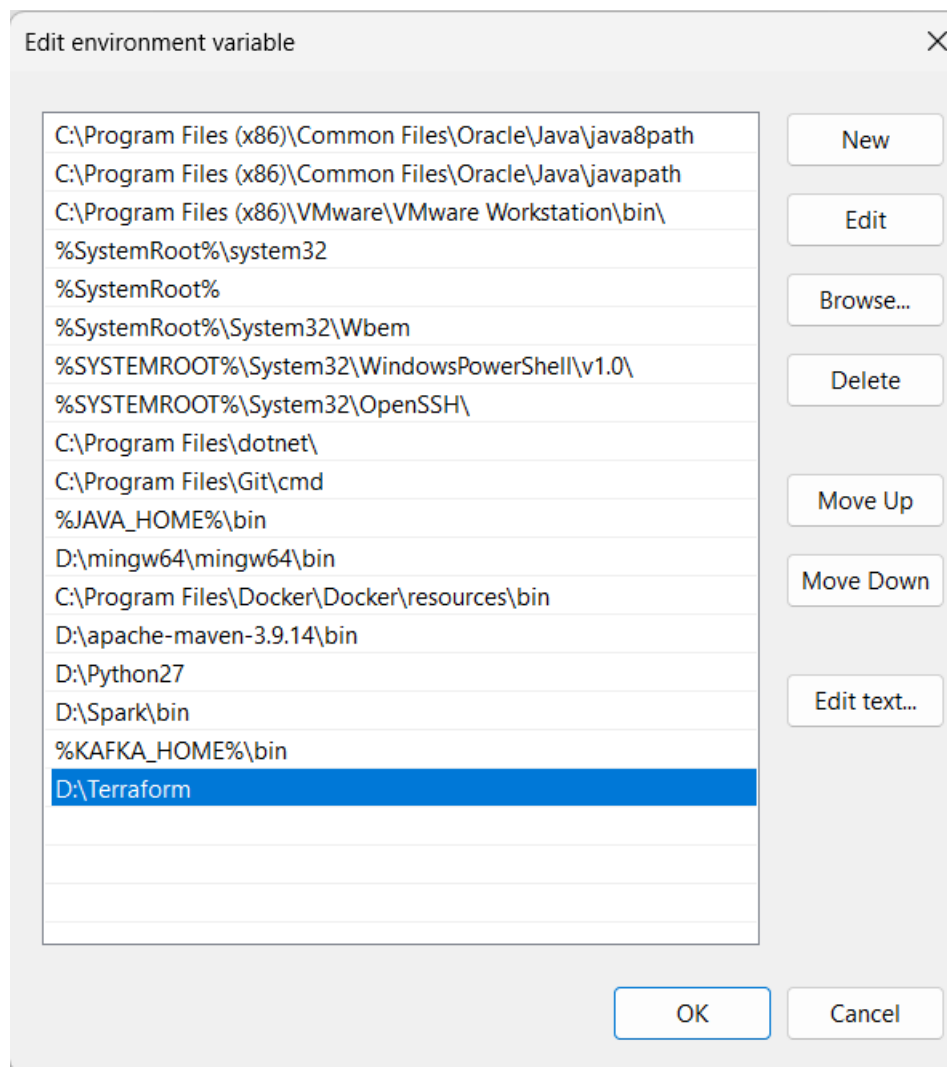


Terraform Mastery Part-1

1. Terraform Local Resource Lifecycle and State Management



Extract it to your desired destination



```
Windows PowerShell
PS C:\Users\sreev> terraform version
Terraform v1.16.3
on windows_amd64
PS C:\Users\sreev> |
```

```
Windows PowerShell
PS D:\Terraform-Lab> code main.tf
PS D:\Terraform-Lab> terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/local versions matching "~> 2.5"...
- Installing hashicorp/local v2.9.1...
- Installed hashicorp/local v2.9.1 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
File Edit Selection View Go Run Terminal Help
main.tf
D:\Terraform-Lab> main.tf
1 terraform{
2   required_providers {
3     local = {
4       source = "hashicorp/local"
5       version = "~> 2.5"
6     }
7   }
8 }
9 resource "local_file" "example"{
10   filename = "vishnu.txt"
11   content = "hi from Vish"
12 }
```

```
Windows PowerShell
PS D:\Terraform-Lab> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# local_file.example will be created
+ resource "local_file" "example" {
+   content            = "hi from Vish"
+   content_base64sha256 = (known after apply)
+   content_base64sha512 = (known after apply)
+   content_md5        = (known after apply)
+   content_sha1        = (known after apply)
+   content_sha256      = (known after apply)
+   content_sha512      = (known after apply)
+   directory_permission = "0777"
+   file_permission     = "0777"
+   filename            = "vishnu.txt"
+   id                  = (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if
you run "terraform apply" now.
PS D:\Terraform-Lab> terraform apply
```

```
Windows PowerShell
PS D:\Terraform-Lab> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# local_file.example will be created
+ resource "local_file" "example" {
+   content            = "hi from Vish"
+   content_base64sha256 = (known after apply)
+   content_base64sha512 = (known after apply)
+   content_md5        = (known after apply)
+   content_sha1        = (known after apply)
+   content_sha256      = (known after apply)
+   content_sha512      = (known after apply)
+   directory_permission = "0777"
+   file_permission     = "0777"
+   filename            = "vishnu.txt"
+   id                  = (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes
```

```
Windows PowerShell

+ content_md5      = (known after apply)
+ content_sha1     = (known after apply)
+ content_sha256   = (known after apply)
+ content_sha512   = (known after apply)
+ directory_permission = "0777"
+ file_permission  = "0777"
+ filename         = "vishnu.txt"
+ id              = (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

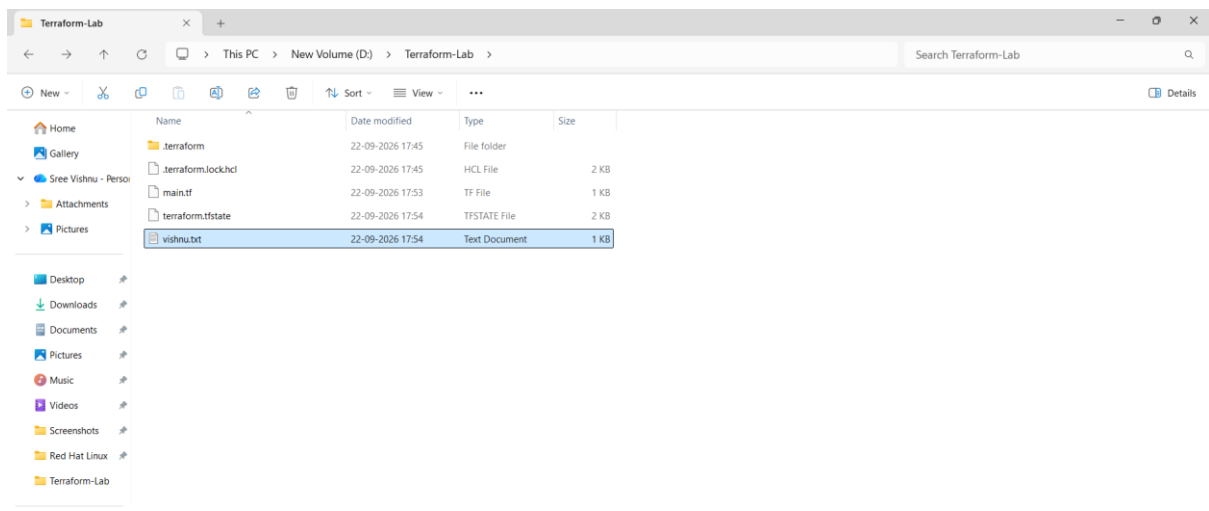
  Enter a value: yes

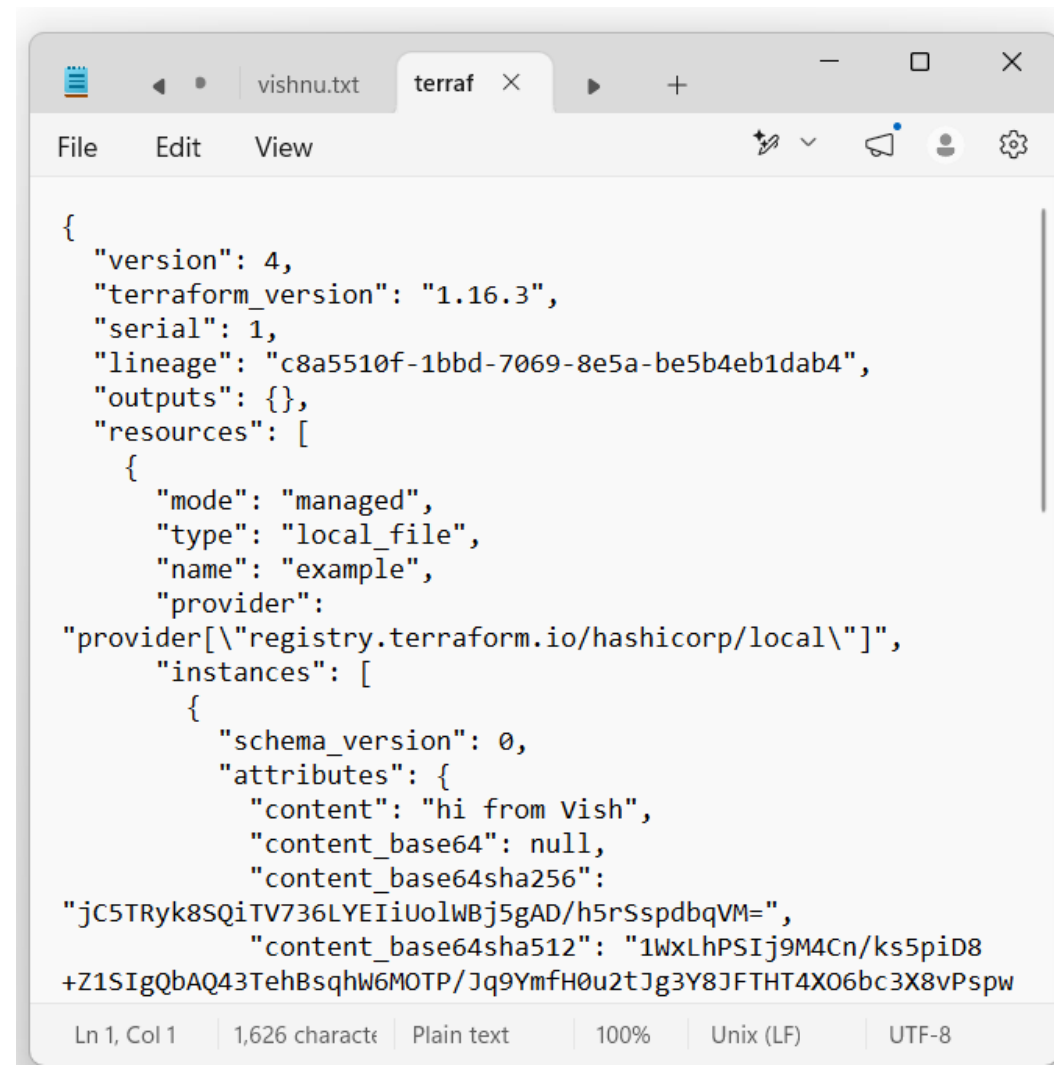
local_file.example: Creating...
local_file.example: Creation complete after 0s [id=c6f080b9cc0441fdb2e7ac8307ecf2cb290703b1]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS D:\Terraform-Lab> terraform plan
local_file.example: Refreshing state... [id=c6f080b9cc0441fdb2e7ac8307ecf2cb290703b1]

No changes. Your infrastructure matches the configuration.

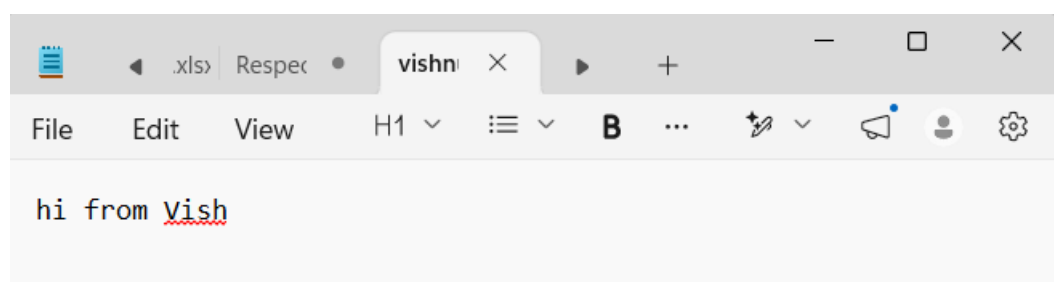
Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are
needed.
PS D:\Terraform-Lab> |
```





```
{
  "version": 4,
  "terraform_version": "1.16.3",
  "serial": 1,
  "lineage": "c8a5510f-1bbd-7069-8e5a-be5b4eb1dab4",
  "outputs": {},
  "resources": [
    {
      "mode": "managed",
      "type": "local_file",
      "name": "example",
      "provider":
"provider[\"registry.terraform.io/hashicorp/local\"]",
      "instances": [
        {
          "schema_version": 0,
          "attributes": {
            "content": "hi from Vish",
            "content_base64": null,
            "content_base64sha256":
"jC5TRYk8SQiTV736LYEIIiUolWBj5gAD/h5rSspdbqVM=",
            "content_base64sha512": "1WxLhPSIj9M4Cn/ks5piD8
+Z1SIgQbAQ43TehBsqrhW6MOTP/Jq9YmfH0u2tJg3Y8JFTHT4X06bc3X8vPspw
```

Ln 1, Col 1 | 1,626 charact | Plain text | 100% | Unix (LF) | UTF-8



```
hi from Vish
```

```
Windows PowerShell
needed.
PS D:\Terraform-Lab> terraform destroy
local_file.example: Refreshing state... [id=c6f080b9cc0441fdb2e7ac8307ecf2cb290703b1]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
- destroy

Terraform will perform the following actions:

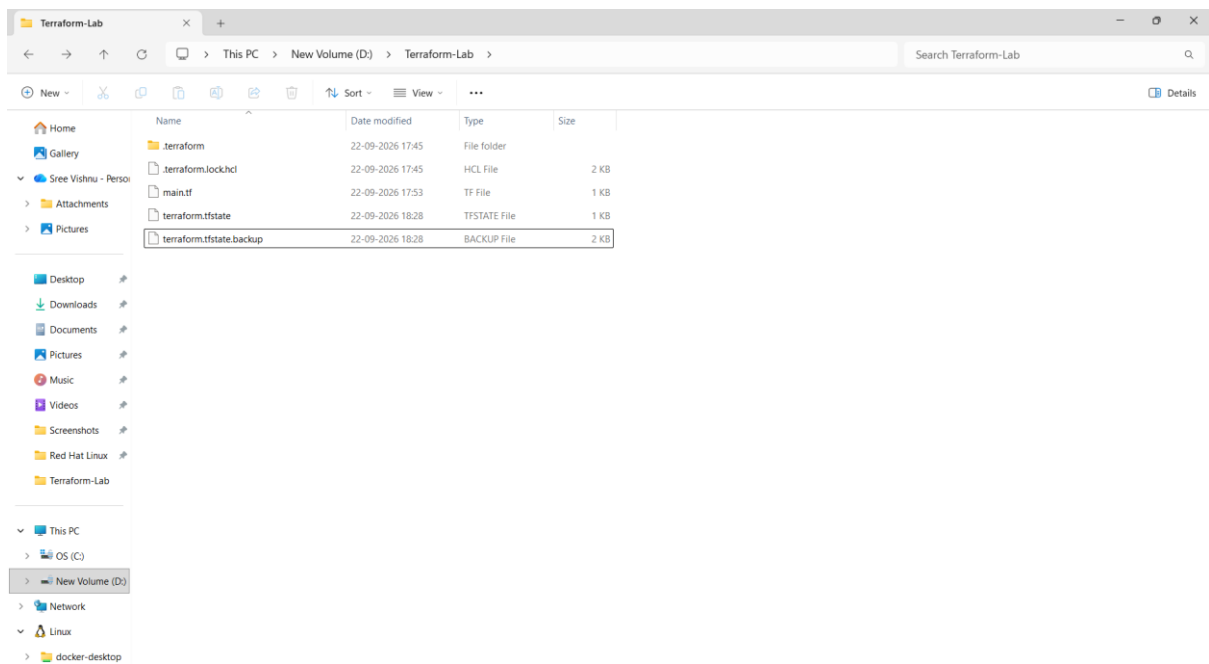
# local_file.example will be destroyed
- resource "local_file" "example" {
  - content          = "hi from Vish" -> null
  - content_base64sha256 = "jC5TRyk8SQiTV736LYEIIuolWBj5gAD/h5rSspdbqVM=" -> null
  - content_base64sha512 = "1WxLhPSIj9M4Cn/ks5piD8+Z1SIgQbAQ43TehBsqhW6MOTP/Jq9YmfH0u2tJg3Y8JFTHT4X06bc3X8vPspwpDg==" -> null
  - content_md5        = "a7d92ead69ae91e65df742f006943711" -> null
  - content_shal       = "c6f080b9cc0441fdb2e7ac8307ecf2cb290703b1" -> null
  - content_sha256     = "8c2e5347293c49089357bdfa2d8108894a255818f98000ff879ad2b2975ba953" -> null
  - content_sha512     = "d56c4b84f4888fd3380a7fe4b39a620fcf99d5222041b010e374de841b2a856e8c3933ff26af5899f1f4bb6b4983763c2454c74f85cee9b7375fcbcfb29c290e" -> null
  - directory_permission = "0777" -> null
  - file_permission     = "0777" -> null
  - filename            = "vishnu.txt" -> null
  - id                  = "c6f080b9cc0441fdb2e7ac8307ecf2cb290703b1" -> null
}

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?
```

```
Enter a value: yes
local_file.example: Destroying... [id=c6f080b9cc0441fdb2e7ac8307ecf2cb290703b1]
local_file.example: Destruction complete after 0s

Destroy complete! Resources: 1 destroyed.
PS D:\Terraform-Lab> |
```



2. Terraform AWS EC2 Provisioning

```
Windows PowerShell
PS C:\Users\sreev> where.exe aws
C:\Users\sreev\AppData\Local\Programs\Amazon\AWSCLIV2\aws.exe
PS C:\Users\sreev> aws --version
aws-cli/2.37.0 Python/3.14.6 Windows/11 script-exe/AMD64
PS C:\Users\sreev> |
```

Note: For installation of AWS CLI you can use the command,

“irm <https://awscli.amazonaws.com/v2/install.ps1> | iex”

The screenshot shows the AWS IAM console interface for creating a new user. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/users/create`. The navigation bar includes the AWS logo, a search bar, and the user profile 'Vishnu@0407 (368995587946)'. The main content area is titled 'Specify user details' and shows a progress indicator with three steps: 'Specify user details' (active), 'Set permissions', and 'Review and create'. The 'User details' section contains a text input field for 'User name' with the value 'tf-admin'. Below this, there is a checkbox labeled 'Provide user access to the AWS Management Console - optional' which is currently unchecked. A note below the checkbox states: 'In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.' At the bottom right, there are 'Cancel' and 'Next' buttons.

IAM > IAM users > Create user

tf-local-dev user group created.

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Specify user details

User details

User name

tf-admin

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☒ Provide user access to the AWS Management Console - optional

In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☒ Autogenerated password

You can view the password after you create the user.

☐ Custom password

Enter a custom password for the user.

☐ Show password

☒ Users must create a new password at next sign-in - Recommended

Users automatically get the `IAMUserChangePassword` policy to allow them to change their own password.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

IAM > IAM users > Create user

tf-local-dev user group created.

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (2)

Search

< 1 >

☐	Group name	Users	Attached policies	Created
☐	AWSSESendingGroupDoNotRename	1	-	2026-08-26 (27 days ago)
☐	tf-local-dev	0	SignInLocalDevelopmentAccess	2026-09-23 (1 minute ago)

Create group

Set permissions boundary - optional

Cancel

Previous

Next

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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Terms

Cookie preferences

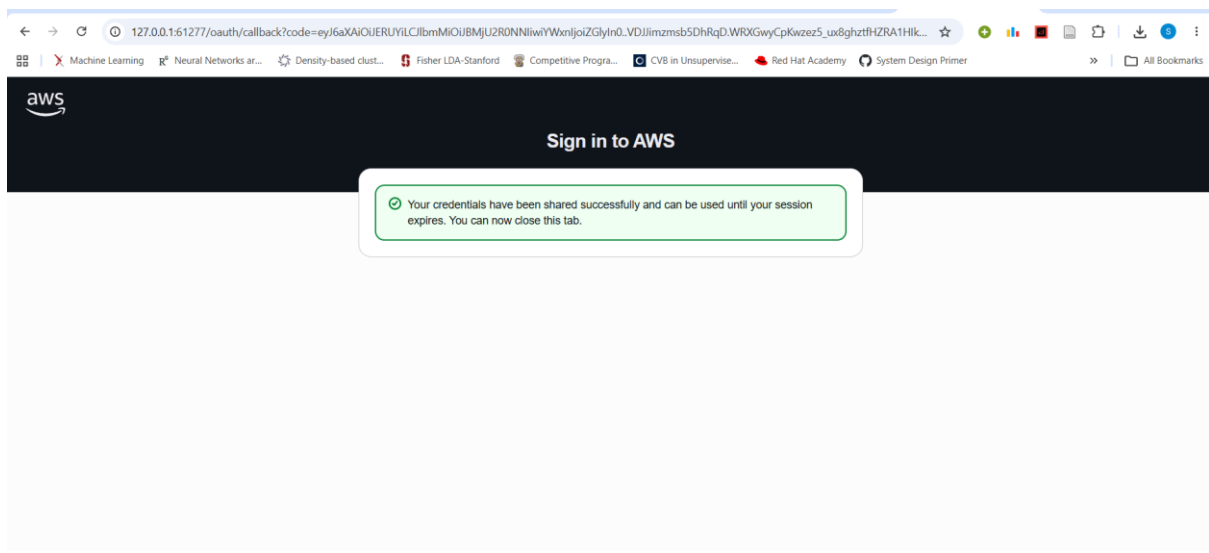

```
Windows PowerShell
PS C:\Users\sreev> where.exe aws
C:\Users\sreev\AppData\Local\Programs\Amazon\AWSCLIV2\aws.exe
PS C:\Users\sreev> aws --version
aws-cli/2.37.0 Python/3.14.6 Windows/11 script-exe/AMD64
PS C:\Users\sreev> aws login
No AWS region has been configured. The AWS region is the geographic location of your AWS resources.

If you have used AWS before and already have resources in your account, specify which region they were created in. If you have not created resources in your account before, you can pick the region closest to you: https://docs.aws.amazon.com/global-infrastructure/latest/regions/aws-regions.html.

You are able to change the region in the CLI at any time with the command "aws configure set region NEW_REGION".
AWS Region [us-east-1]: ap-south-1
Attempting to open your default browser. If the browser does not open, open the following URL.
If you are unable to open the URL on this device, run this command again with the '--remote' option.

https://ap-south-1.signin.aws.amazon.com/v1/authorize?response_type=code&client_id=arn%3Aaws%3Asignin%3A%3A%3Adevtools%2Fsame-device&state=193ad16d-078d-4f07-8a60-fd763e4d9c17&code_challenge_method=SHA-256&scope=openid&redirect_uri=http%3A%2F%2F127.0.0.1%3A61277%2Foauth%2Fcallback&code_challenge=T-5j7rXAu_2nBXadQ3eYtkMDE_6hNhN0kH8eJmJBzdI

Updated profile default to use arn:aws:iam::368999587946:user/tf-admin credentials.
PS C:\Users\sreev> |
```



```
Windows PowerShell
PS C:\Users\sreev> where.exe aws
C:\Users\sreev\AppData\Local\Programs\Amazon\AWSCLI\aws.exe
PS C:\Users\sreev> aws --version
aws-cli/2.37.0 Python/3.14.6 Windows/11 script-exe/AMD64
PS C:\Users\sreev> aws login
No AWS region has been configured. The AWS region is the geographic location of your AWS resources.

If you have used AWS before and already have resources in your account, specify which region they were created in. If you have not created resources in your account before, you can pick the region closest to you: https://docs.aws.amazon.com/global-infrastructure/latest/regions/aws-regions.html.

You are able to change the region in the CLI at any time with the command "aws configure set region NEW_REGION".
AWS Region [us-east-1]: ap-south-1
Attempting to open your default browser. If the browser does not open, open the following URL.
If you are unable to open the URL on this device, run this command again with the '--remote' option.

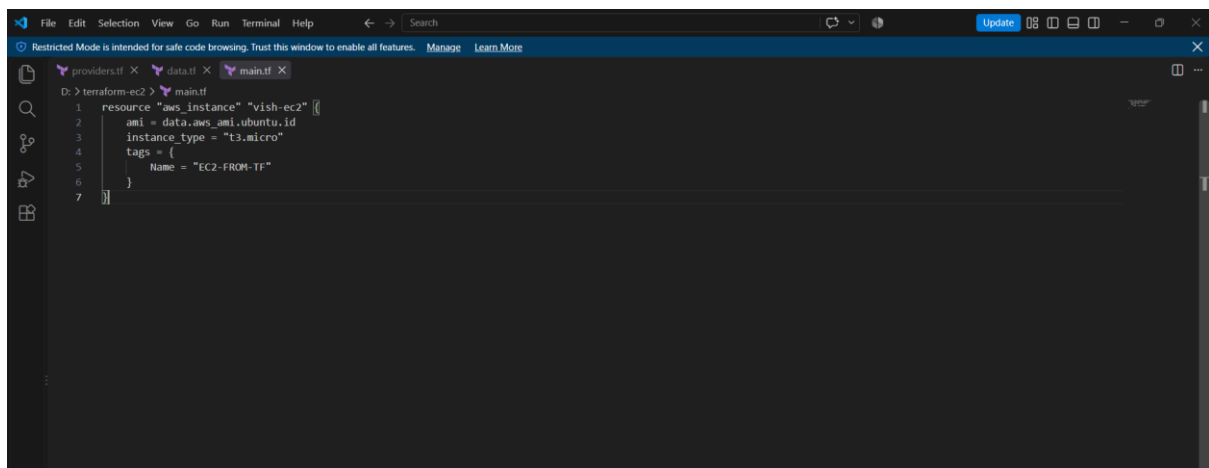
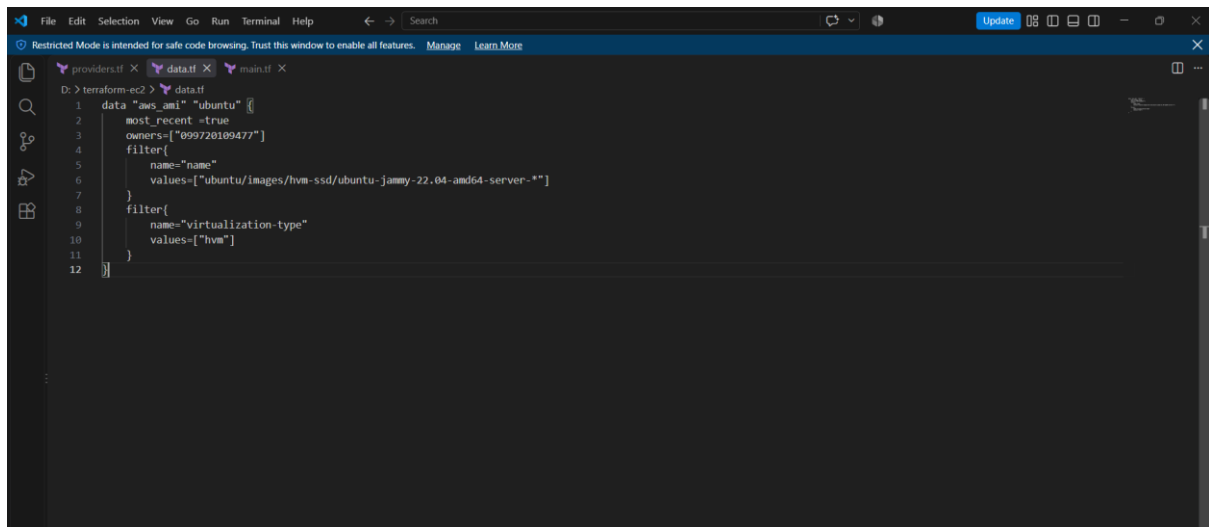
https://ap-south-1.signin.aws.amazon.com/v1/authorize?response_type=code&client_id=arn%3Aaws%3Asignin%3A%3A%3Adevtools%2Fsame-device&state=193ad16d-078d-4f07-8a60-fd763e4d9c17&code_challenge_method=SHA-256&scope=openid&redirect_uri=http%3A%2F%2F127.0.0.1%3A61277%2Foauth%2Fcallback&code_challenge=T-5j7rXAu_2nBXadQ3eYtkMDE_6hNhN0kH8eJmJBzdI

Updated profile default to use arn:aws:iam::368999587946:user/tf-admin credentials.
PS C:\Users\sreev> aws sts get-caller-identity
{
  "UserId": "AIDAVL2QW4BVFVVDQON2",
  "Account": "368999587946",
  "Arn": "arn:aws:iam::368999587946:user/tf-admin"
}

PS C:\Users\sreev> |
```

Note: AWS STS is the service/mechanism involved in issuing temporary credentials; those credentials are then used to authenticate requests to AWS services

```
File Edit Selection View Go Run Terminal Help
D:\> terraform-ec2 > providers.tf
providers.tf data.tf main.tf
1 terraform{
2   required_providers{
3     aws={
4       source = "hashicorp/aws"
5       version = "~> 6.0"
6     }
7   }
8 }
9 provider "aws" {
10   region = "ap-south-1"
11 }
```



```
PS D:\terraform-ec2> terraform init
Initializing the backend...

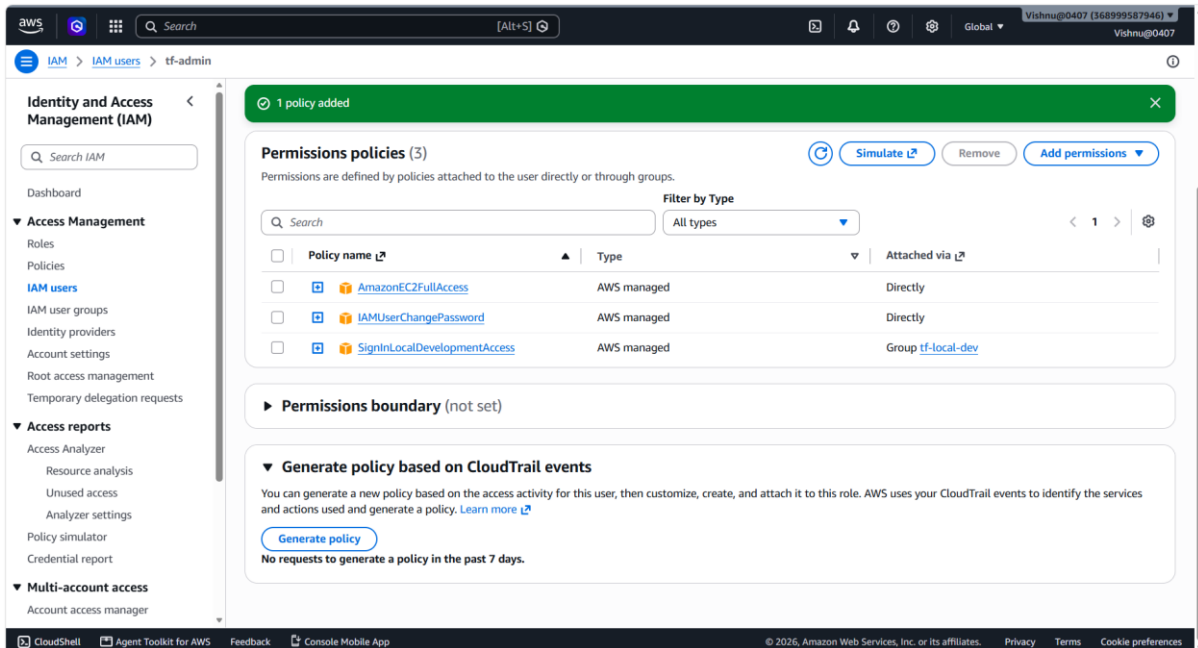
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.66.0...
- Installed hashicorp/aws v6.66.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```



```
Windows PowerShell
PS D:\terraform-ec2> terraform plan
data.aws_ami.ubuntu: Reading...
data.aws_ami.ubuntu: Read complete after 1s [id=ami-07dfe2c11d33a069a]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.vish-ec2 will be created
+ resource "aws_instance" "vish-ec2" {
  + ami                  = "ami-07dfe2c11d33a069a"
  + arn                  = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone      = (known after apply)
  + disable_api_stop       = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized          = (known after apply)
  + enable_primary_ipv6     = (known after apply)
  + force_destroy          = false
  + get_password_data       = false
  + host_id               = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile    = (known after apply)
  + id                    = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle      = (known after apply)
  + instance_state          = (known after apply)
  + instance_type           = "t3.micro"
```

```
Windows PowerShell

+ public_ip                = (known after apply)
+ region                  = "ap-south-1"
+ secondary_private_ips    = (known after apply)
+ security_groups          = (known after apply)
+ source_dest_check        = true
+ spot_instance_request_id = (known after apply)
+ subnet_id               = (known after apply)
+ tags                    = {
+   + "Name" = "EC2-FROM-TF"
+ }
+ tags_all                = {
+   + "Name" = "EC2-FROM-TF"
+ }
+ tenancy                 = (known after apply)
+ user_data_base64        = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids  = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)
```

```
Windows PowerShell

- tags_all                = {} -> null
- throughput              = 0 -> null
- volume_id               = "vol-0b1a43c7991b917bb" -> null
- volume_size             = 8 -> null
- volume_type             = "gp2" -> null
  # (1 unchanged attribute hidden)
}

~ secondary_network_interface (known after apply)
}

Plan: 1 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.vish-ec2: Destroying... [id=i-0f48bc55b6417fdf7]
aws_instance.vish-ec2: Still destroying... [id=i-0f48bc55b6417fdf7, 00m10s elapsed]
aws_instance.vish-ec2: Still destroying... [id=i-0f48bc55b6417fdf7, 00m20s elapsed]
aws_instance.vish-ec2: Still destroying... [id=i-0f48bc55b6417fdf7, 00m30s elapsed]
aws_instance.vish-ec2: Destruction complete after 31s
aws_instance.vish-ec2: Creating...
aws_instance.vish-ec2: Still creating... [00m10s elapsed]
aws_instance.vish-ec2: Creation complete after 14s [id=i-0f5ee588c59b58177]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
PS D:\terraform-ec2>
```

Instance summary for i-0f5ee588c59b58177 (EC2-FROM-TF) Info

Updated less than a minute ago

Instance ID i-0f5ee588c59b58177	Public IPv4 address 43.204.147.103 open address	Private IPv4 addresses 172.31.4.100
IPv6 address -	Instance state Running	Public DNS ec2-43-204-147-103.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-4-100.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-4-100.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding ⚠ User: arn:aws:iam::368999587946:user/tf-admin is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action Retry
Auto-assigned IP address 43.204.147.103 [Public IP]	VPC ID vpc-02b3dda78c3050d	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0bcb3c45270b8df08	Managed false
IMDSv2 Optional	Instance ARN arn:aws:ec2:ap-south-1:368999587946:instance/i-0f5ee588c59b58177	

⚠ EC2 recommends setting IMDSv2 to required.

```

PS D:\terraform-ec2> terraform destroy
data.aws_ami.ubuntu: Reading...
data.aws_ami.ubuntu: Read complete after 6s [id=ami-07dfe2c11d33a069a]
aws_instance.vish-ec2: Refreshing state... [id=i-0f5ee588c59b58177]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  - destroy

Terraform will perform the following actions:

# aws_instance.vish-ec2 will be destroyed
- resource "aws_instance" "vish-ec2" {
  - ami              = "ami-07dfe2c11d33a069a" -> null
  - arn              = "arn:aws:ec2:ap-south-1:368999587946:instance/i-0f5ee588c59b58177" -> nul
}

- associate_public_ip_address = true -> null
- availability_zone            = "ap-south-1b" -> null
- disable_api_stop             = false -> null
- disable_api_termination     = false -> null
- ebs_optimized                = false -> null
- force_destroy                = false -> null
- get_password_data            = false -> null
- hibernation                  = false -> null
- id                           = "i-0f5ee588c59b58177" -> null
- instance_initiated_shutdown_behavior = "stop" -> null
- instance_state               = "running" -> null
- instance_type                = "t3.micro" -> null
- ipv6_address_count           = 0 -> null
- ipv6_addresses               = [] -> null
  
```

```
Windows PowerShell X + v
- delete_on_termination = true -> null
- device_name            = "/dev/sda1" -> null
- encrypted              = false -> null
- iops                   = 100 -> null
- tags                   = {} -> null
- tags_all               = {} -> null
- throughput             = 0 -> null
- volume_id              = "vol-053fec75fe0883ec4" -> null
- volume_size            = 8 -> null
- volume_type            = "gp2" -> null
  # (1 unchanged attribute hidden)
}
}

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

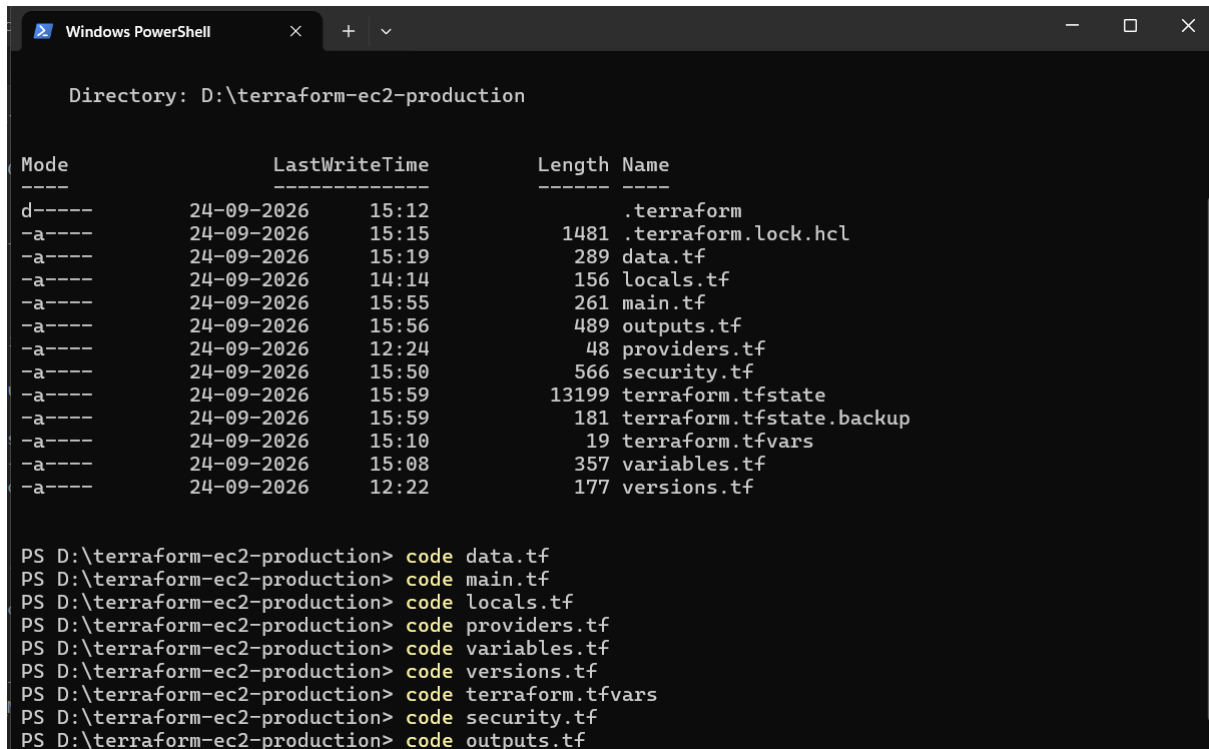
aws_instance.vish-ec2: Destroying... [id=i-0f5ee588c59b58177]
aws_instance.vish-ec2: Still destroying... [id=i-0f5ee588c59b58177, 00m10s elapsed]
aws_instance.vish-ec2: Still destroying... [id=i-0f5ee588c59b58177, 00m20s elapsed]
aws_instance.vish-ec2: Still destroying... [id=i-0f5ee588c59b58177, 00m30s elapsed]
aws_instance.vish-ec2: Destruction complete after 31s

Destroy complete! Resources: 1 destroyed.
PS D:\terraform-ec2>
```

The screenshot displays the AWS Management Console interface. At the top, the navigation bar shows the AWS logo, a search bar, and the user's profile (Vishnu@0407 (368999587946) tf-admin). The left-hand navigation pane lists various services and categories, with 'Instances' selected under the 'EC2' category. The main content area shows the 'Instance summary' for the instance 'i-0f5ee588c59b58177' (EC2-FROM-TF). The instance is in a 'Terminated' state. The summary includes details such as Public IPv4 address, Instance type (t3.micro), VPC ID, Subnet ID, and IAM role. A warning message indicates that the user is not authorized to perform certain actions due to an identity-based policy.

Instance summary for i-0f5ee588c59b58177 (EC2-FROM-TF)		
Updated less than a minute ago		
Instance ID i-0f5ee588c59b58177	Public IPv4 address -	Private IPv4 addresses -
IPv6 address -	Instance state Terminated	Public DNS -
Hostname type -	Instance type t3.micro	Elastic IP addresses -
Answer private resource DNS name -	VPC ID -	AWS Compute Optimizer finding User: arn:aws:iam::368999587946:user/tf-admin is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action Retry
Auto-assigned IP address -	Subnet ID -	Auto Scaling Group name -
IAM role -	Instance ARN arn:aws:ec2:ap-south-1:368999587946:instance/i-0f5ee588c59b58177	Managed false

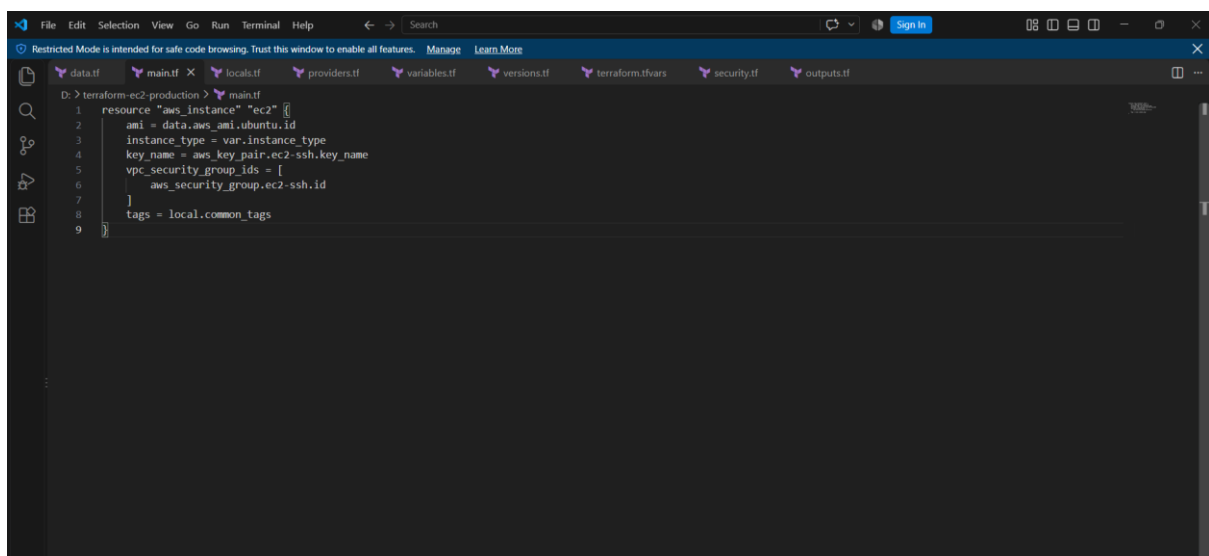
3. Parameterized AWS Infrastructure Using Terraform Variables, Outputs, Locals and Data Sources



Directory: D:\terraform-ec2-production

Mode	LastWriteTime	Length	Name
d-----	24-09-2026 15:12		.terraform
-a----	24-09-2026 15:15	1481	.terraform.lock.hcl
-a----	24-09-2026 15:19	289	data.tf
-a----	24-09-2026 14:14	156	locals.tf
-a----	24-09-2026 15:55	261	main.tf
-a----	24-09-2026 15:56	489	outputs.tf
-a----	24-09-2026 12:24	48	providers.tf
-a----	24-09-2026 15:50	566	security.tf
-a----	24-09-2026 15:59	13199	terraform.tfstate
-a----	24-09-2026 15:59	181	terraform.tfstate.backup
-a----	24-09-2026 15:10	19	terraform.tfvars
-a----	24-09-2026 15:08	357	variables.tf
-a----	24-09-2026 12:22	177	versions.tf

```
PS D:\terraform-ec2-production> code data.tf
PS D:\terraform-ec2-production> code main.tf
PS D:\terraform-ec2-production> code locals.tf
PS D:\terraform-ec2-production> code providers.tf
PS D:\terraform-ec2-production> code variables.tf
PS D:\terraform-ec2-production> code versions.tf
PS D:\terraform-ec2-production> code terraform.tfvars
PS D:\terraform-ec2-production> code security.tf
PS D:\terraform-ec2-production> code outputs.tf
```



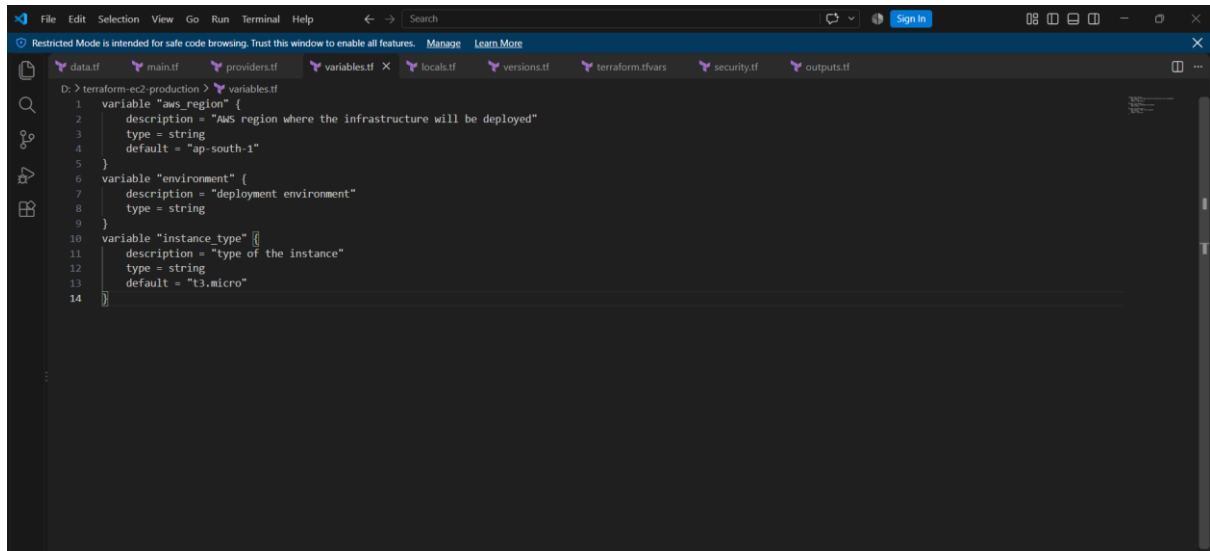
```
D:\> terraform-ec2-production > main.tf
1 resource "aws_instance" "ec2" {
2   ami = data.aws_ami.ubuntu.id
3   instance_type = var.instance_type
4   key_name = aws_key_pair.ec2-ssh.key_name
5   vpc_security_group_ids = [
6     aws_security_group.ec2-ssh.id
7   ]
8   tags = local.common_tags
9 }
```

The screenshot shows a code editor window with a dark theme. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. Below the menu bar, there is a status bar indicating "Restricted Mode is intended for safe code browsing. Trust this window to enable all features." and a "Sign In" button. The editor displays a file named "data.tf" with the following Terraform configuration:

```
D: > terraform-ec2-production > data.tf
1 data "aws_ami" "ubuntu" {
2   most_recent = true
3   owners      = ["099720109477"]
4   filter {
5     name   = "name"
6     values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
7   }
8   filter {
9     name   = "virtualization-type"
10    values = ["hvm"]
11  }
12 }
```

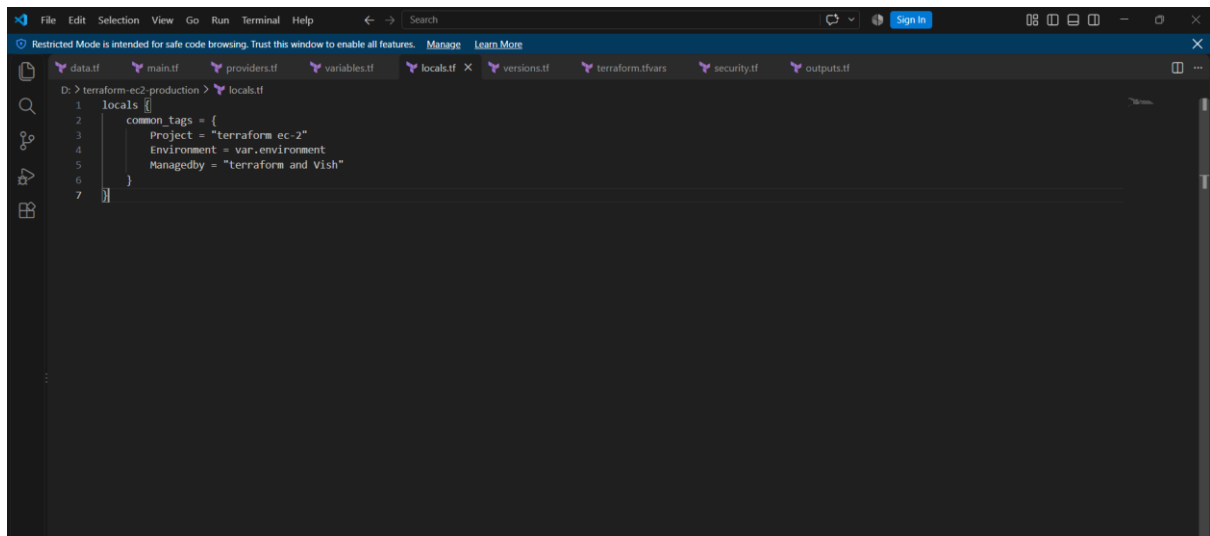
The screenshot shows a code editor window with a dark theme. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. Below the menu bar, there is a status bar indicating "Restricted Mode is intended for safe code browsing. Trust this window to enable all features." and a "Sign In" button. The editor displays a file named "providers.tf" with the following Terraform configuration:

```
D: > terraform-ec2-production > providers.tf
1 provider "aws" {
2   region = var.aws_region
3 }
```



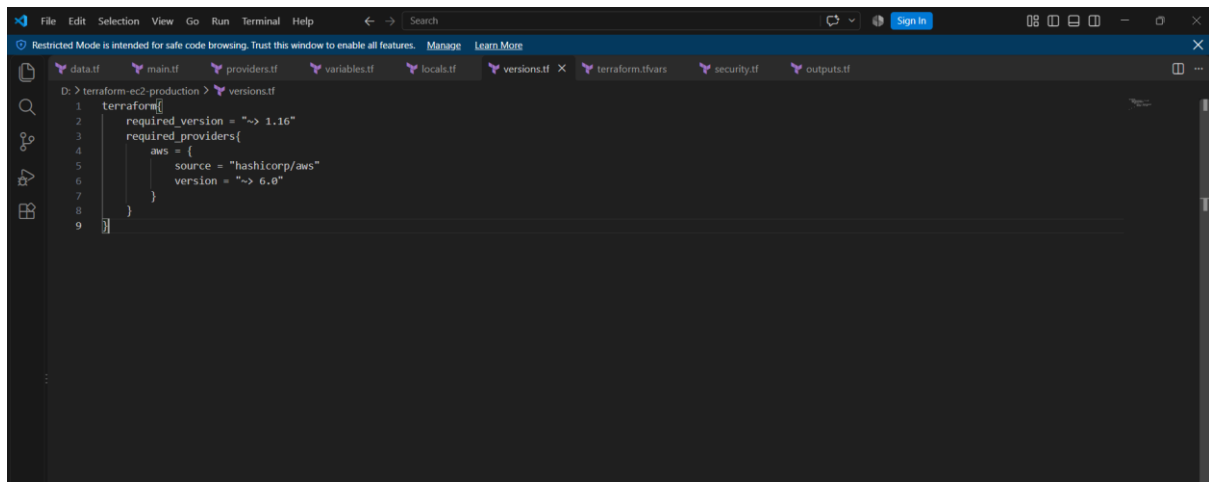
This screenshot shows the Visual Studio Code editor with the file `variables.tf` open. The editor displays three Terraform variable definitions. The first variable is `aws_region`, a string type with a description of "AWS region where the infrastructure will be deployed" and a default value of "ap-south-1". The second variable is `environment`, also a string type, with a description of "deployment environment". The third variable is `instance_type`, a string type, with a description of "type of the instance" and a default value of "t3.micro". The file explorer on the left shows the project structure with files like `data.tf`, `main.tf`, `providers.tf`, `variables.tf`, `locals.tf`, `versions.tf`, `terraform.tfvars`, `security.tf`, and `outputs.tf`.

```
1 variable "aws_region" {
2   description = "AWS region where the infrastructure will be deployed"
3   type = string
4   default = "ap-south-1"
5 }
6
7 variable "environment" {
8   description = "deployment environment"
9   type = string
10 }
11
12 variable "instance_type" {
13   description = "type of the instance"
14   type = string
15   default = "t3.micro"
16 }
```

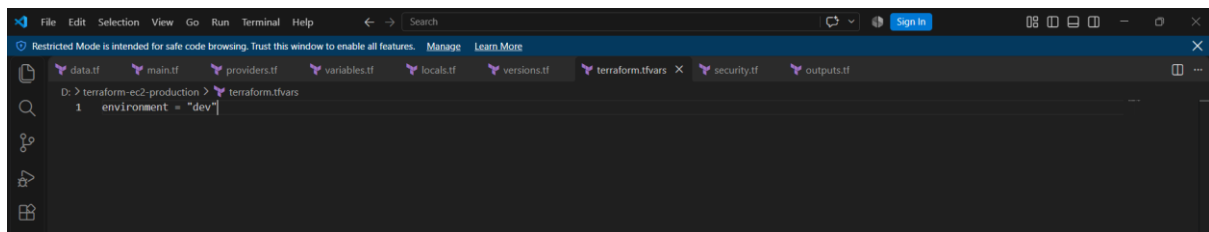


This screenshot shows the Visual Studio Code editor with the file `locals.tf` open. The editor displays a Terraform `locals` block containing a `common_tags` map. The map includes three entries: `Project` with the value "terraform ec-2", `Environment` with the value `var.environment`, and `Managedby` with the value "terraform and Vish". The file explorer on the left shows the project structure with files like `data.tf`, `main.tf`, `providers.tf`, `variables.tf`, `locals.tf`, `versions.tf`, `terraform.tfvars`, `security.tf`, and `outputs.tf`.

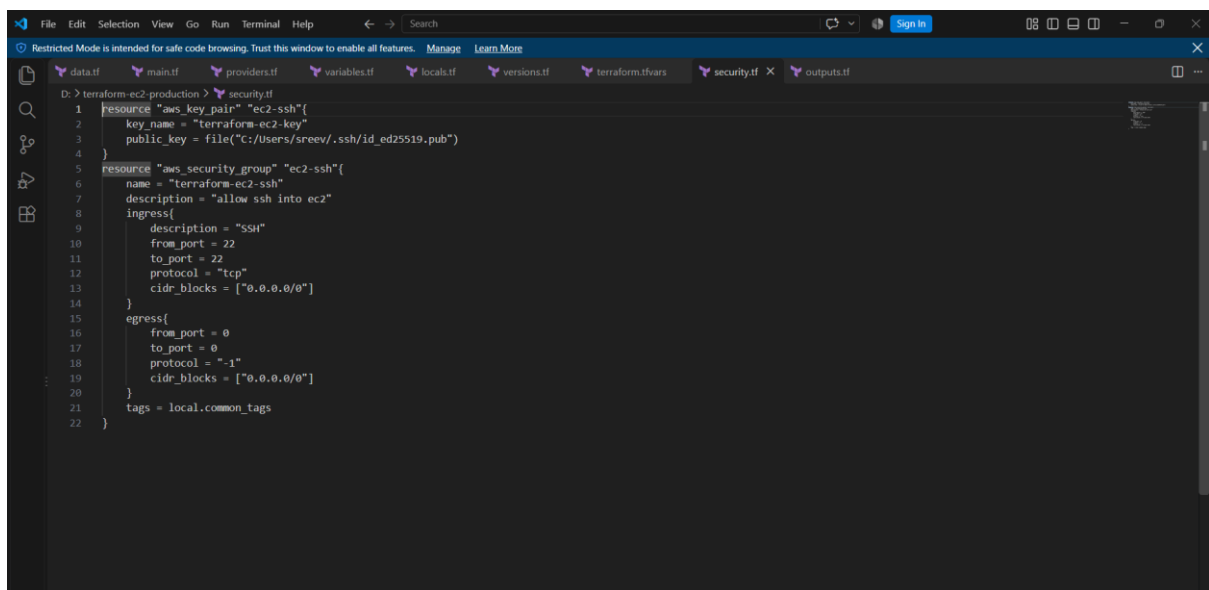
```
1 locals {
2   common_tags = {
3     Project = "terraform ec-2"
4     Environment = var.environment
5     Managedby = "terraform and Vish"
6   }
7 }
```



```
1 terraform{
2   required_version = "~> 1.16"
3   required_providers{
4     aws = {
5       source = "hashicorp/aws"
6       version = "~> 6.0"
7     }
8   }
9 }
```

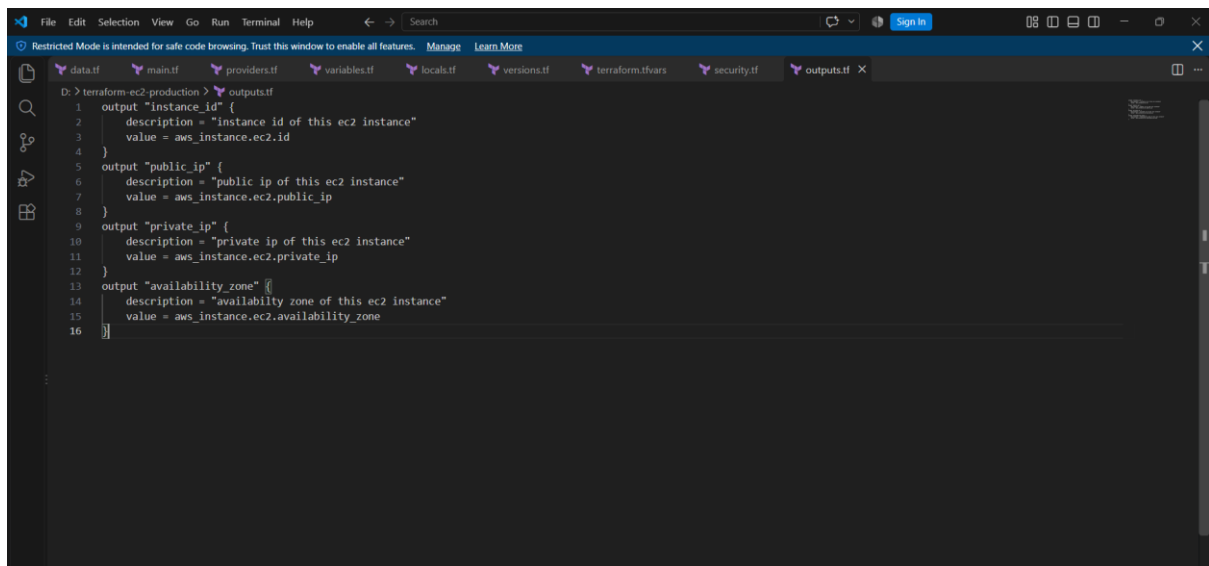


```
1 environment = "dev"
```



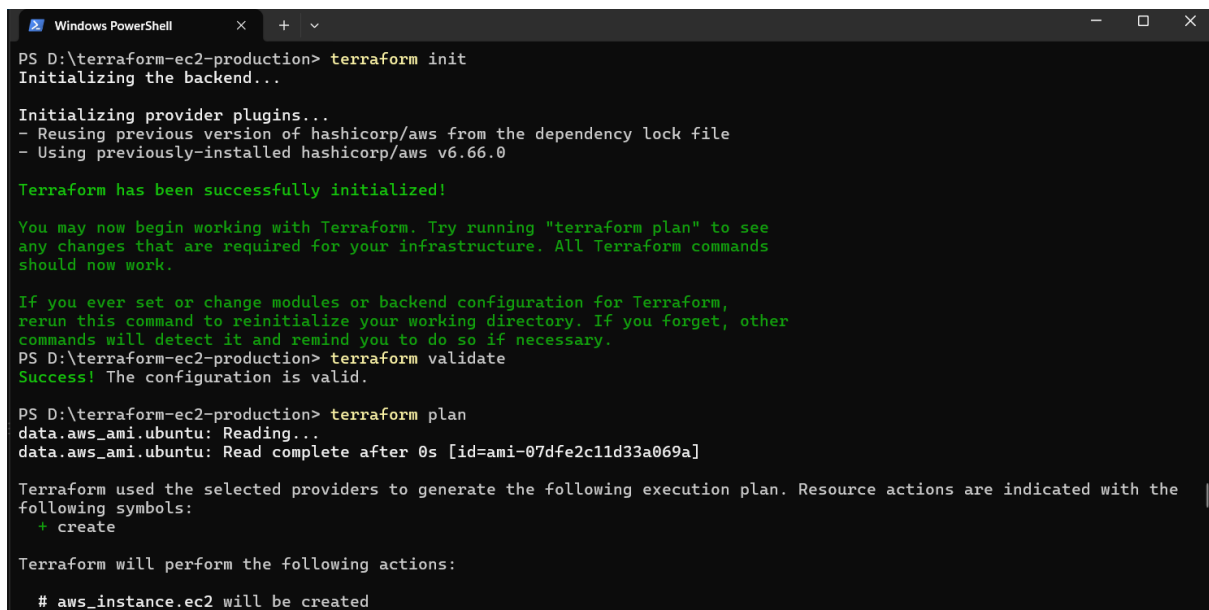
```
1 resource "aws_key_pair" "ec2-ssh"{
2   key_name = "terraform-ec2-key"
3   public_key = file("C:/Users/sreev/.ssh/id_ed25519.pub")
4 }
5 resource "aws_security_group" "ec2-ssh"{
6   name = "terraform-ec2-ssh"
7   description = "allow ssh into ec2"
8   ingress{
9     description = "SSH"
10    from_port = 22
11    to_port = 22
12    protocol = "tcp"
13    cidr_blocks = ["0.0.0.0/0"]
14  }
15  egress{
16    from_port = 0
17    to_port = 0
18    protocol = "-1"
19    cidr_blocks = ["0.0.0.0/0"]
20  }
21  tags = local.common_tags
22 }
```

****Main****



The screenshot shows a code editor with a dark theme. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. Below the menu bar, there's a search bar and a 'Sign in' button. The editor displays several files in the left sidebar: data.tf, main.tf, providers.tf, variables.tf, locals.tf, versions.tf, terraform.tfvars, security.tf, and outputs.tf. The main editor window shows the contents of 'outputs.tf', which contains Terraform output definitions for instance_id, public_ip, private_ip, and availability_zone. The code is as follows:

```
1 output "instance_id" {
2   description = "instance id of this ec2 instance"
3   value = aws_instance.ec2.id
4 }
5 output "public_ip" {
6   description = "public ip of this ec2 instance"
7   value = aws_instance.ec2.public_ip
8 }
9 output "private_ip" {
10  description = "private ip of this ec2 instance"
11  value = aws_instance.ec2.private_ip
12 }
13 output "availability_zone" {
14  description = "availability zone of this ec2 instance"
15  value = aws_instance.ec2.availability_zone
16 }
```



The screenshot shows a Windows PowerShell terminal window with the following output:

```
PS D:\terraform-ec2-production> terraform init
Initializing the backend...

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.66.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS D:\terraform-ec2-production> terraform validate
Success! The configuration is valid.

PS D:\terraform-ec2-production> terraform plan
data.aws_ami.ubuntu: Reading...
data.aws_ami.ubuntu: Read complete after 0s [id=ami-07dfe2c11d33a069a]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.ec2 will be created
```

```
Windows PowerShell
PS D:\terraform-ec2-production> terraform plan
data.aws_ami.ubuntu: Reading...
data.aws_ami.ubuntu: Read complete after 0s [id=ami-07dfe2c11d33a069a]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.ec2 will be created
+ resource "aws_instance" "ec2" {
  + ami                  = "ami-07dfe2c11d33a069a"
  + arn                  = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone     = (known after apply)
  + disable_api_stop      = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized         = (known after apply)
  + enable_primary_ipv6    = (known after apply)
  + force_destroy         = false
  + get_password_data      = false
  + host_id               = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile   = (known after apply)
  + id                   = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle     = (known after apply)
  + instance_state         = (known after apply)
  + instance_type          = "t3.micro"
  + outpost_arn            = (known after apply)
  + password_data          = (known after apply)
  + placement_group        = (known after apply)
  + placement_group_id     = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns            = (known after apply)
  + private_ip             = (known after apply)
  + public_dns             = (known after apply)
  + public_ip              = (known after apply)
  + region                 = "ap-south-1"
  + secondary_private_ips   = (known after apply)
  + security_groups         = (known after apply)
  + source_dest_check       = true
  + spot_instance_request_id = (known after apply)
  + subnet_id              = (known after apply)
  + tags                   = {
    + "Environment" = "dev"
    + "Managedby"  = "terraform and Vish"
    + "Project"     = "terraform ec-2"
  }
  + tags_all            = {
    + "Environment" = "dev"
    + "Managedby"  = "terraform and Vish"
    + "Project"     = "terraform ec-2"
  }
  + tenancy              = (known after apply)
  + user_data_base64     = (known after apply)
  + user_data_replace_on_change = false
  + vpc_security_group_ids = (known after apply)
}
```

```
Windows PowerShell
+ outpost_arn            = (known after apply)
+ password_data          = (known after apply)
+ placement_group        = (known after apply)
+ placement_group_id     = (known after apply)
+ placement_partition_number = (known after apply)
+ primary_network_interface_id = (known after apply)
+ private_dns            = (known after apply)
+ private_ip             = (known after apply)
+ public_dns             = (known after apply)
+ public_ip              = (known after apply)
+ region                 = "ap-south-1"
+ secondary_private_ips   = (known after apply)
+ security_groups         = (known after apply)
+ source_dest_check       = true
+ spot_instance_request_id = (known after apply)
+ subnet_id              = (known after apply)
+ tags                   = {
  + "Environment" = "dev"
  + "Managedby"  = "terraform and Vish"
  + "Project"     = "terraform ec-2"
}
+ tags_all            = {
  + "Environment" = "dev"
  + "Managedby"  = "terraform and Vish"
  + "Project"     = "terraform ec-2"
}
+ tenancy              = (known after apply)
+ user_data_base64     = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
```

```
Windows PowerShell
# aws_key_pair.ec2-ssh will be created
+ resource "aws_key_pair" "ec2-ssh" {
+   arn                = (known after apply)
+   fingerprint       = (known after apply)
+   id                = (known after apply)
+   key_name           = "terraform-ec2-key"
+   key_name_prefix    = (known after apply)
+   key_pair_id        = (known after apply)
+   key_type           = (known after apply)
+   public_key         = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIFb5hV6YSNU3IGc3s0EzZsScV4DTpDso9tRHrWXEDBmT sreev@admin"
+   region             = "ap-south-1"
+   tags_all           = (known after apply)
+ }

# aws_security_group.ec2-ssh will be created
+ resource "aws_security_group" "ec2-ssh" {
+   arn                = (known after apply)
+   description        = "allow ssh into ec2"
+   egress              = [
+     {
+       cidr_blocks     = [
+         "0.0.0.0/0",
+       ]
+       from_port        = 0
+       ipv6_cidr_blocks = []
+       prefix_list_ids  = []
+       protocol         = "-1"
+       security_groups  = []
+       self              = false
+       to_port          = 0
+     }
+   ]
+ }
```

```
Windows PowerShell
    ]
+   description        = "SSH"
+   from_port          = 22
+   ipv6_cidr_blocks   = []
+   prefix_list_ids    = []
+   protocol           = "tcp"
+   security_groups     = []
+   self               = false
+   to_port            = 22
+ },
+ ]
+ name                 = "terraform-ec2-ssh"
+ name_prefix          = (known after apply)
+ owner_id             = (known after apply)
+ region              = "ap-south-1"
+ revoke_rules_on_delete = false
+ tags                 = {
+   "Environment" = "dev"
+   "Managedby"  = "terraform and Vish"
+   "Project"     = "terraform ec-2"
+ }
+ tags_all             = {
+   "Environment" = "dev"
+   "Managedby"  = "terraform and Vish"
+   "Project"     = "terraform ec-2"
+ }
+ vpc_id              = (known after apply)
+ }

Plan: 3 to add, 0 to change, 0 to destroy.
```

```
Windows PowerShell
+ name_prefix      = (known after apply)
+ owner_id         = (known after apply)
+ region           = "ap-south-1"
+ revoke_rules_on_delete = false
+ tags             = {
+   "Environment" = "dev"
+   "Managedby"  = "terraform and Vish"
+   "Project"     = "terraform ec-2"
+ }
+ tags_all         = {
+   "Environment" = "dev"
+   "Managedby"  = "terraform and Vish"
+   "Project"     = "terraform ec-2"
+ }
+ vpc_id           = (known after apply)
}

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ availability_zone = (known after apply)
+ instance_id      = (known after apply)
+ private_ip       = (known after apply)
+ public_ip        = (known after apply)

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if
you run "terraform apply" now.
PS D:\terraform-ec2-production> |
```

```
Windows PowerShell

Changes to Outputs:
+ availability_zone = (known after apply)
+ instance_id      = (known after apply)
+ private_ip       = (known after apply)
+ public_ip        = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_key_pair.ec2-ssh: Creating...
aws_security_group.ec2-ssh: Creating...
aws_key_pair.ec2-ssh: Creation complete after 1s [id=terraform-ec2-key]
aws_security_group.ec2-ssh: Creation complete after 3s [id=sg-0e3a4c399fb2baafc]
aws_instance.ec2: Creating...
aws_instance.ec2: Still creating... [00m10s elapsed]
aws_instance.ec2: Creation complete after 13s [id=i-04223144e88807f88]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

availability_zone = "ap-south-1b"
instance_id      = "i-04223144e88807f88"
private_ip       = "172.31.3.171"
public_ip        = "3.6.160.244"
PS D:\terraform-ec2-production> |
```

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-04223144e88807f88

Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer

aws Search [Alt+S] Asia Pacific (Mumbai) Vishnu@0407 (36899587946) tf-admin

EC2 > Instances > i-04223144e88807f88

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- ▼ Images
 - AMIs
 - AMI Catalog
- ▼ Elastic Block Store
 - Volumes
 - Snapshots
 - Lifecycle Manager

Instance summary for i-04223144e88807f88

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-04223144e88807f88	Public IPv4 address 3.6.160.244 open address	Private IPv4 addresses 172.31.3.171
IPv6 address -	Instance state Running	Public DNS ec2-3-6-160-244.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-3-171.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-3-171.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding User: arnaws:iam::36899587946:user/tf-admin is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action Retry
Auto-assigned IP address 3.6.160.244 [Public IP]	VPC ID vpc-02b3dda78cff3050d	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0bcb3c45270b8df08	Managed false
IMDSv2 Optional	Instance ARN arnaws:ec2:ap-south-1:36899587946:instance/i-04223144e88807f88	

EC2 recommends getting IMDSv2 to required I

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

aws Search [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Vishnu@0407 (36899587946) tf-admin

EC2 > Instances > i-04223144e88807f88 > Connect to instance

Instance i-04223144e88807f88 State: Running Status: 0/2 checks passed	VPC vpc-02b3dda78cff3050d	Subnet subnet-0bcb3c45270b8df08	Security groups sg-0e3a4c399fb2baafc (terraform-ec2-ssh)
---	-------------------------------------	---	--

In web browser [In SSH client](#)

SSH info

[Connect assist](#)

SSH key pair assigned at launch terraform-ec2-key	Public DNS ec2-3-6-160-244.ap-south-1.compute.amazonaws.com	Username ubuntu
---	---	---------------------------

How to connect

- Open an SSH client
- Locate or download your private key file
- Ensure your key is not publicly viewable
- Connect to your instance using its Public DNS

```
chmod 400 "terraform-ec2-key.pem"
```

ssh -i "terraform-ec2-key.pem" ubuntu@ec2-3-6-160-244.ap-south-1.compute.amazonaws.com

Copied command

```
ubuntu@ip-172-31-3-171: ~
sreev@admin MINGW64 ~/.ssh (main)
$ ls
id_ed25519 id_ed25519.pub id_rsa id_rsa.pub known_hosts known_hosts.old
sreev@admin MINGW64 ~/.ssh (main)
$ ssh -i id_ed25519 ubuntu@ec2-3-6-160-244.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-3-6-160-244.ap-south-1.compute.amazonaws.com (3.6.160.244)' can't be established.
ED25519 key fingerprint is SHA256:M5H2vJLejKykaEa3sJAzgVzsHNvAKqLokpZNabBguDk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-6-160-244.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1063-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 24 10:32:59 UTC 2026

System load:  0.0                Processes:    112
Usage of /:   23.0% of 7.57GB    Users logged in: 0
```

```
ubuntu@ip-172-31-3-171: ~
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-3-171:~$ |
```

There we go!! Successfully connected via SSH